

MING-CHIEH, HU

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Education

Delft University of Technology (TU Delft)

MSc Geomatics

MSc Thesis: *Gaussian Splatting to Piecewise-Planar Surfaces for 3D Building Reconstruction*

Delft, Netherlands

Sep. 2024 — present

National Yang Ming Chiao Tung University (NYCU)

B.S. in Electronics Engineering

Innovative Technology and Art Program

Hsinchu, Taiwan

Sep. 2017 — Jun. 2021

Sep. 2019 — Jun. 2021

Experience

TU Delft

Teaching Assistant

Delft, Netherlands

Sep. 2025 – present

- **Providing assistance in four courses**, namely, *Python Programming*, *Digital Terrain Modelling*, *Positioning and Location Awareness*, and *3D Modelling*.

Readar

Intern Developer

Utrecht, Netherlands

Jul. 2025 – Oct. 2025

- Proposed a hybrid method to **solve a zonal statistics problem** that is 4.5 - 7× faster than baseline for a post-processing bottleneck, with reasonable memory consumption.
- Developed a **GUI and CLI tool for data quality control** and inspection (footprint calculation and misalignment detection). This tool utilizes optimal file formats and parallel processing methods to replace the obsolete script.
- Benchmarked the performance of modern raster file formats (GeoTIFF/COG/Zarr) for random-access tasks in cloud environment.

TORIII Technology

Project Manager / Full Stack Developer / UI Designer

Taipei, Taiwan

Sep. 2022 – Nov. 2023

- Planned, designed, and developed a **customer relationship management system** that served thousands of users. Including features such as monthly charges, copy-trade automation, and a dashboard, using Figma, Node.js, React, MongoDB and GCP.
- Planned and coordinated a customized development of a **ticketing platform**. Worked with designers and engineers in a project manager role.
- Collaborated with Aiii, an AI marketing company, to develop an **NFT membership management** system and a centralized wallet that operates on Ethereum and Polygon chains.
- Developed a quant-trading strategy optimization method with an concise user interface.

Solar Decathlon Europe 21/22

Student Electrical Engineer, team TDIS

Wuppertal, Germany

Sep. 2020 – Jul. 2022

- Designed and implemented an **electrical plan** for a two story house demonstration unit, integrating photovoltaic panels, inverters and batteries using Grasshopper 3D.
- Led the **transportation research of Taipei** and innovative design for urban mobility. Engaged in discussions with the juries, articulating our design and strategies.
- Awarded 1_{st} Prize for architecture, 3_{rd} Prize for innovation among 18 international teams.

Learning-based Voice Conversion

Member of the Artificial Intelligence and Multimedia Laboratory

Hsinchu, Taiwan

Sep. 2019 – Jun. 2020

- Enhanced the voice conversion results of a Variational Auto Encoder by integrating MOSNet and subsequently yielded a new loss metric.

Skills

- **Relevant Coursework:** Algorithms, Data Structures, OOP, Machine Learning, Calculus I-II, Probability and Statistics, Linear Algebra, Sensing Technologies, GIS and Cartography, Geo Database Management Systems, Digital Terrain Modelling, Photogrammetry and 3D Computer Vision, Wind Modelling
- **Programming:** C, C++, JavaScript, Python, HTML, CSS, SQL
- **Libraries:** PyTorch, Express, React, CGAL, GDAL, Rasterio, Zarr, GeoParquet
- **Software:** QGIS, GRASS GIS, FME, Rhinoceros 3D, Grasshopper 3D, Git, GCP, MongoDB, Oracle Database, PostgreSQL, PostGIS, IfcOpenShell
- **TOEFL:** 102/120